TPG4190 Radon and F-K filters

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Overview

- F-K filters
- The Radon transform
- Radon demultiple

2D Fourier Transform

$$p(x,t) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} dk_x \int_{-\infty}^{+\infty} d\omega \, P(k_x,\omega) \exp[i(\omega t - k_x x)], \quad (1)$$

$$P(k_x,\omega) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dt \, p(x,t) \exp[-i(\omega t - k_x x)]. \quad (2)$$

2D Fourier Transform

- 2D Discrete Fourier Transform

$$P_{m,n} = \frac{1}{NM} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} p_{k,l} \exp(-2\pi i m l / M) \exp(2\pi i n k / N) \quad (3)$$

- $p_{k,l}$ is the sampling of $P(x,t)$

$$p_{k,l} = P(k\Delta x, l\Delta t), \text{ where } k = 0, 1, \ldots, M \text{ and } l = 0, 1, \ldots, N. \quad (4)$$

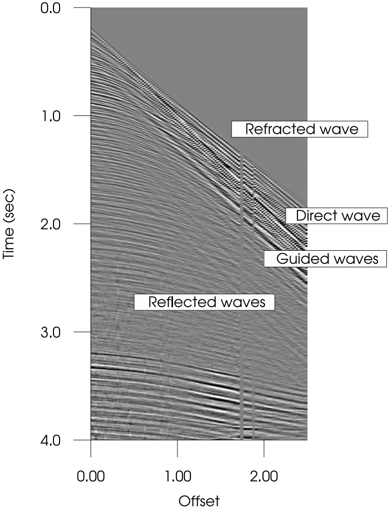
The inverse 2D Discrete Fourier transform is

$$p_{k,l} = \frac{1}{NM} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} P_{m,n} \exp(2\pi i m l / M) \exp(-2\pi i n k / N). \quad (5)$$

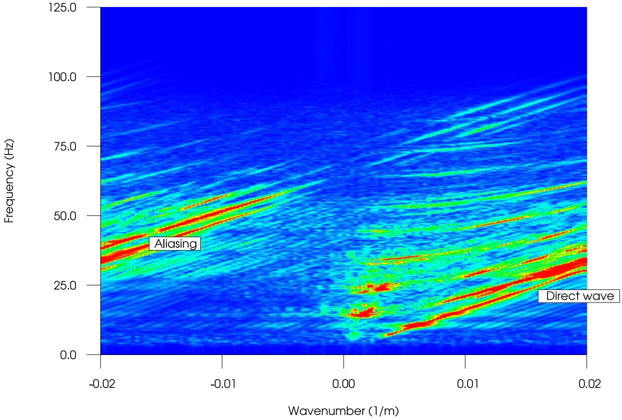
F-K Spectra

- Identify Noise patterns
- Aliasing

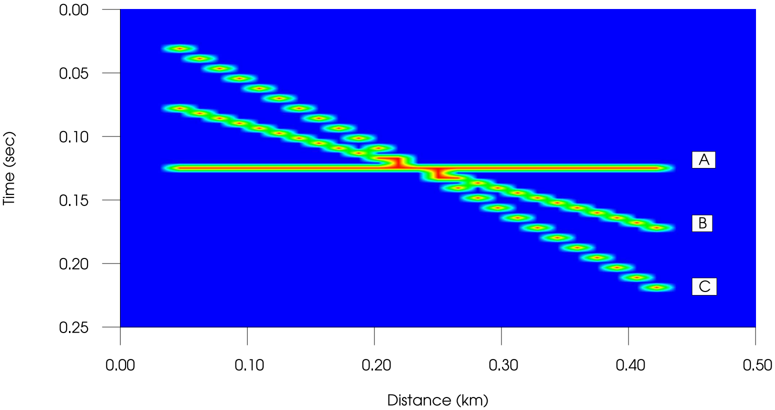
F-K spectra


0.0
1.0
2.0
3.0
4.0
Time (sec)
0.00
1.00
2.00
Offset
Refracted wave
Direct wave
Guided waves
Reflected waves

F-K Filters


125.0
100.0
75.0
50.0
25.0
0.0
Frequency (Hz)
Aliasing
Direct wave
-0.02
-0.01
0.00
0.01
0.02
Wavenumber (1/m)

F-K Filters



0.00
0.05
0.10
0.15
0.20
0.25
Time (sec)
A
B
C
0.00
0.10
0.20
0.30
0.40
0.50
Distance (km)

F-K Filters

How does a straight line in the x-t domain map to the f-k domain? A linear event in the x-t domain can easily be described using a δ-function

$$\delta(t - x/c_x - t_0), \tag{6}$$

where c_x is the (apparent) velocity in the x-direction and t_0 is a constant. If we use equation (2) we get

$$\int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dt\, \delta(t - x/c_x - t_0) \exp[-i(\omega t - k_x x)]. \tag{7}$$

F-K Filters

By performing the integral over time we get

$$\int_{-\infty}^{+\infty} dx \, \exp[-i(\omega \frac{x}{c_x} + \omega t_0 - k_x x)]. \tag{8}$$

By rearranging the above expression one gets

$$\int_{-\infty}^{+\infty} dx \, \exp(-i\omega t_0) \exp[-ix(\omega/c_x - k_x)], \tag{9}$$

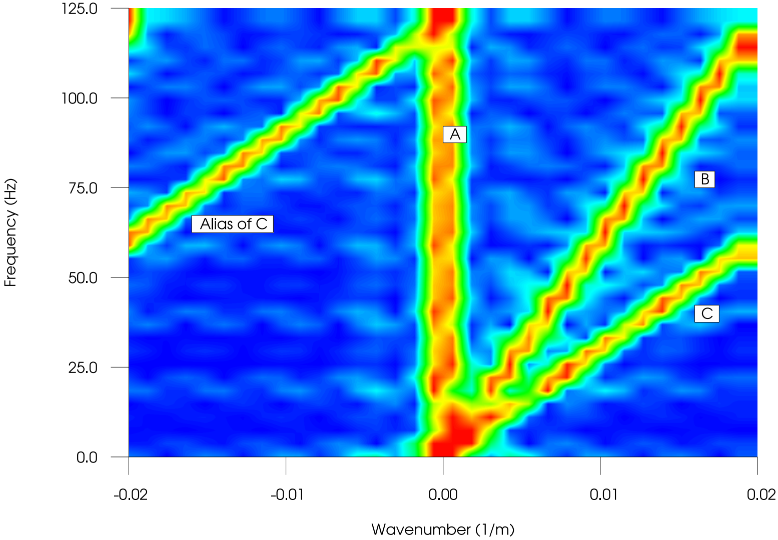
which according to Rottman (1960), page 110, is equal to

$$2\pi \exp(-i\omega t_0)\delta(\omega/c_x - kx). \tag{10}$$

10 implies that the line $t - x/c_x - t_0$ in the x-t domain maps to the line $\omega - c_x k_x$ in the f-k domain.
If we introduce the reduced wavenumber, $\bar{k}_x = k_x/2\pi$, and the frequency $f = \omega/2\pi$, the line in the f-k domain is described by

$$f = c_x \bar{k}_x. \tag{11}$$

F-K Filters



125.0
100.0
75.0
50.0
25.0
0.0
Frequency (Hz)
-0.02
-0.01
0.00
0.01
0.02
Wavenumber (1/m)
A
B
C
Alias of C

F-K Filters

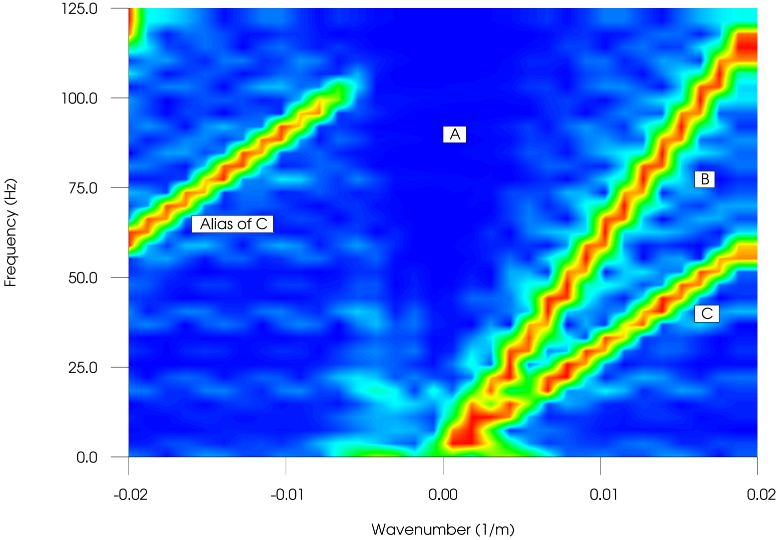
Event A is described by the line

$$x = t_0 = 0.14\,\text{secomd}, \tag{12}$$

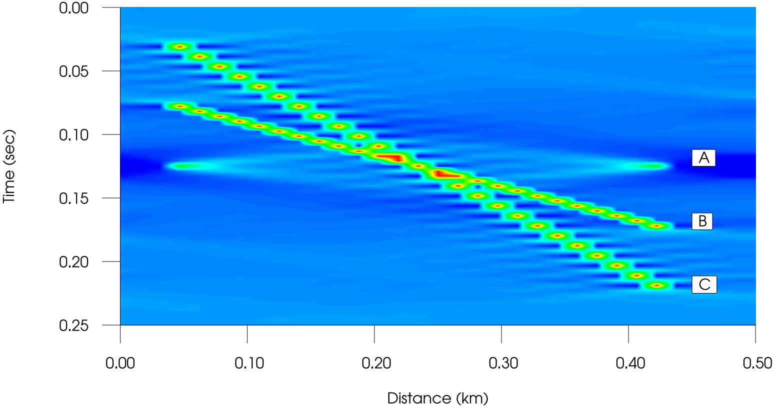
and $c_x = \infty$. The corresponding line in the f-k domain, would be described by the line

$$\bar{k}_x = 0. \tag{13}$$

F-K Filters


125.0
100.0
75.0
50.0
25.0
0.0
Frequency (Hz)
A
B
C
Alias of C
-0.02
-0.01
0.00
0.01
0.02
Wavenumber (1/m)

F-K Filters



0.00
0.05
0.10
0.15
0.20
0.25
Time (sec)
A
B
C
0.00
0.10
0.20
0.30
0.40
0.50
Distance (km)

F-K Filters

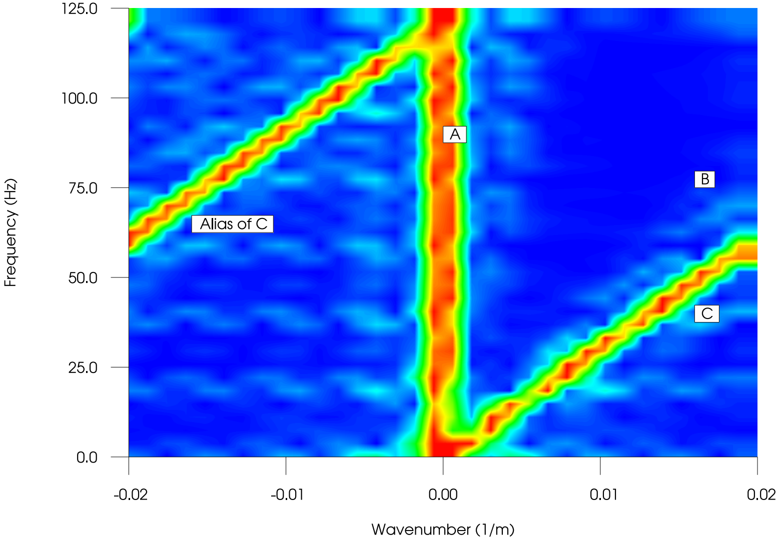
Event B is described by the line

$$t = \frac{x}{c_x} + t_0, \tag{14}$$

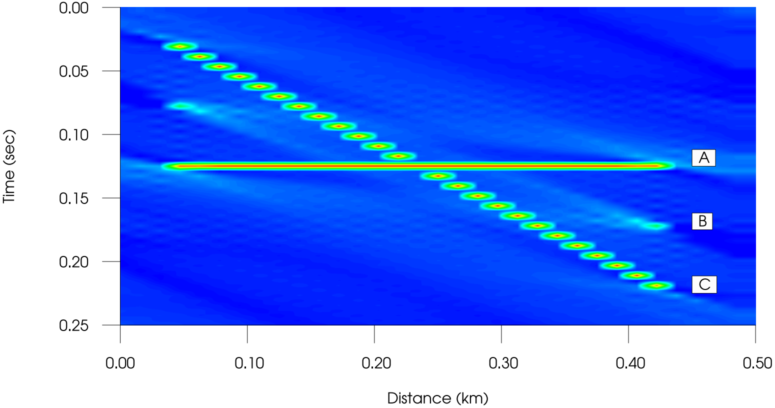
where $c_x = 4000$ and $t_0 = 0.07$ seconds.

$$f = 4000\bar{k}_x, \tag{15}$$

F-K Filters


125.0
100.0
75.0
50.0
25.0
0.0
Frequency (Hz)
A
B
C
Alias of C
-0.02
-0.01
0.00
0.01
0.02
Wavenumber (1/m)

F-K Filters



0.00
0.05
0.10
0.15
0.20
0.25
Time (sec)
0.00
0.10
0.20
0.30
0.40
0.50
Distance (km)
A
B
C

F-K Filters

Event C is spatially aliased, which can be seen from the fact that this event is described by the line

$$t = \frac{x}{2000.0} + 0.025, \tag{16}$$

corresponding to the line in the f-k domain

$$f = 2000\bar{k}. \tag{17}$$

The Nyquist limit is

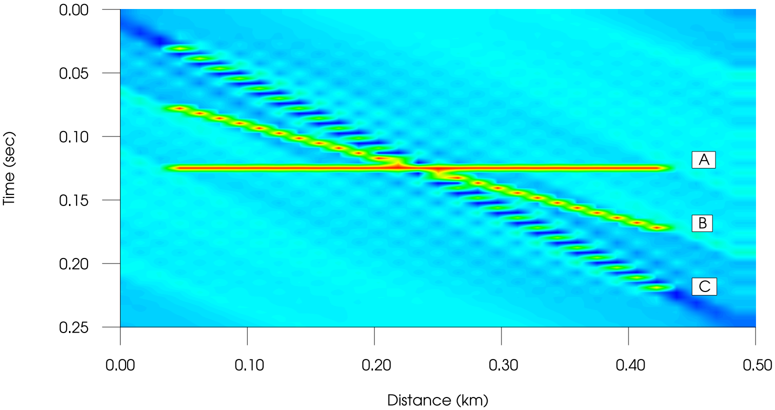
$$k_x = \frac{\omega}{c_x} = \frac{\pi}{\Delta x}, \tag{18}$$

giving

$$f = \frac{c_x}{2\Delta x} = \frac{2000}{32} = 62.5\,\mathrm{Hz}. \tag{19}$$

The maximum frequency in the data is 125Hz, which is larger than 62.5 Hz.

F-K Filters



0.00
0.05
0.10
0.15
0.20
0.25
Time (sec)
A
B
C
0.00
0.10
0.20
0.30
0.40
0.50
Distance (km)

The Radon transform

- 2D Radon transform of a function

$$\hat{f}(p,\tau) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dt\delta(t - px - \tau) f(x,t), \tag{20}$$

- p: slope
- τ : intercept
- integration over t:

$$\hat{f}(p,\tau) = \int_{-\infty}^{+\infty} dx f(x, \tau + px). \tag{21}$$

The Radon transform

- Linear Radon transform or a slant stack
- The line described by

$$t = px + \tau, \tag{22}$$

- is mapped to p, τ in the Radon transformed domain.

The Radon transform

- Fourier transform of equation (21) over τ

$$\hat{f}(p,\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} d\tau \int_{-\infty}^{+\infty} dx f(x,\tau+px)\exp(-i\omega\tau) \tag{23}$$

- Integration variable τ changed to $u = \tau + px$ equation

$$\hat{f}(p,\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} du \int_{-\infty}^{+\infty} dx f(x,u)\exp(-i\omega u + \omega px) \tag{24}$$

- $k_x = \omega p$, one gets

$$\hat{f}(k_x/\omega,\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} du \int dx f(x,u)\exp(-i\omega u + k_x x). \tag{25}$$

The Radon transform

- Inverse transform

$$f(x,t) = -\frac{1}{2\pi^2}\int dp \int d\tau \frac{\partial_\tau \hat{f}(p,\tau - px)}{\tau - t}. \tag{26}$$

The Radon transform of travel-time hyperbolas

The traveltime of a single primary reflection in a CMP-gather is

$$t^2 = t_0^2 + x^2/c^2. \tag{27}$$

c is the traveltime, and t_0 is the zero-offset traveltime. The slownes p is defined as

$$p = \frac{dt}{dx}, \tag{28}$$

The Radon transform of travel-time hyperbolas

which gives by using equation (27)

$$p = \frac{x}{tc^2}. \tag{29}$$

or

$$t = \frac{x}{pc^2}. \tag{30}$$

Inserting equation (30) into equation (27) gives

$$x = \frac{pt_0c^2}{\sqrt{1 - p^2c^2}}. \tag{31}$$

Also inserting equation (31) into (30) gives

$$t = \frac{t_0}{\sqrt{1 - p^2c^2}}. \tag{32}$$

The Radon transform of travel-time hyperbolas

Using equations (32) and (31) we get

$$\tau = t - px = t_0\sqrt{(1 - p^2c^2)}. \tag{33}$$

The last equation automatically gives

$$\left(\frac{\tau}{ct_0}\right)^2 + p^2 = \frac{1}{c^2}, \tag{34}$$

which shows that the two parameters τ and p lies on an ellipse. An hyperbolic traveltime curve in the x, t space is the transformed to an ellipse in the $\tau - p$ space.

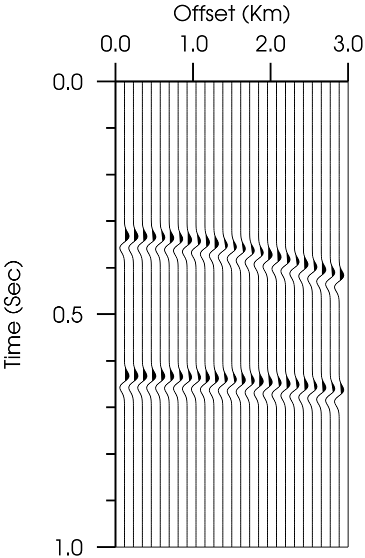


Figure: CMP with two events with velocity equal to 2000 m/s and 2500 m/s.

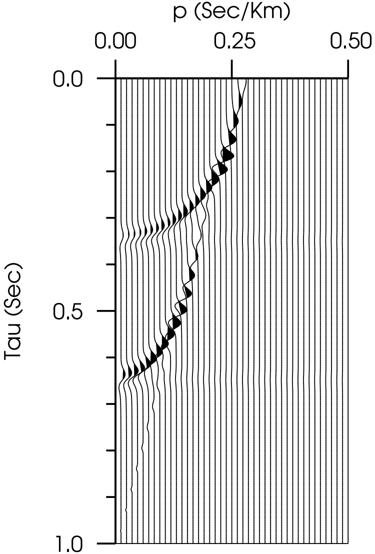


Figure: $\tau - p$ transform of the CMP-gather shown in figure 1

Radon Multiple removal

- Raypaths with multiple reflections
- Multiples are unwanted
- Multiples are usually removed

Principle of Radon demultiple

Traveltime-distance curve for a primary reflection from the bottom of a layer with constant velocity

$$t_p = \sqrt{\tau_0^2 + \frac{4h^2}{c^2}}, \tag{35}$$

and the first multiple reflection from the same reflector

$$t_m = \sqrt{(2\tau_0)^2 + \frac{4(2h)^2}{c^2}}. \tag{36}$$

Principle of Radon demultiple

Consider also a primary reflection arriving from some deeper reflector at the same time

$$t_p = \sqrt{(2\tau_0)^2 + \frac{4(2h)^2}{c_{rms}^2}}. \tag{37}$$

Although arriving at the same time as the multiple reflection, the curvature of this event is different, because the velocity is $c_{rms} > c$.

Principle of Radon demultiple

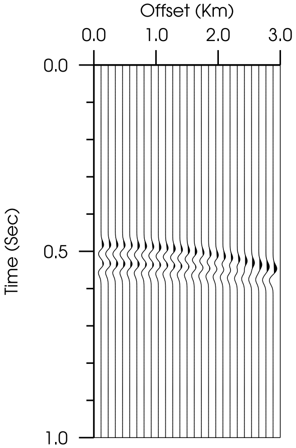


Figure: Cmp with primary reflection and multiple reflection interfering.

Principle of Radon demultiple

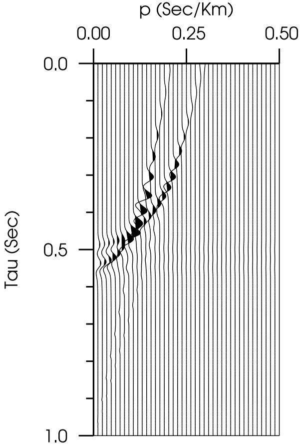


Figure: Cmp with primary reflection and multiple reflection from 3 in the tau-p domain. The apparent velocity for the multiple reflection is lower than the primary reflection, making separation of the two events possible.

Principle of Radon demultiple

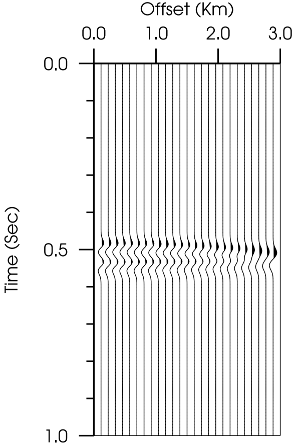


Figure: Cmp with primary reflection and multiple reflection after nmo-correction with a velocity higher than the multiple velocity but lower than the primary velocity.

Principle of Radon demultiple

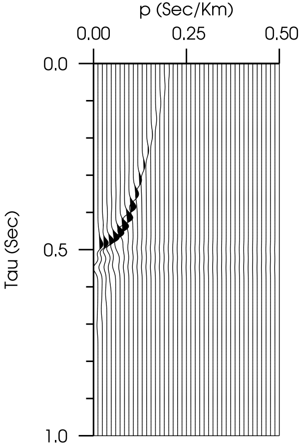


Figure: Cmp with primary reflection and multiple reflection from figure 5 after a *tau* − *p*-p transform. Only the multiple reflection is now visible, the primary appears for small negative p-values (not plotted)

Principle of Radon demultiple

- Straightforward mute and inverse transform is not possible
- Inverse radon gives too many artefacts

Parabolic Radon

$$\hat{f}(p,\tau) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dt \delta(t - px^2 - \tau) f(x,t). \qquad (38)$$

- Transform maps an event in the x-t domain described by a parabola into a point
- Usefull for nmo-corrected data

Parabolic Radon

In general the inverse transform can be considered to be of the general form

$$f(x,t) = \int_{-\infty}^{+\infty} dp \hat{f}(\tau = t - px^2, p), \tag{39}$$

or in discrete form

$$f(x_k,t) = \sum_{l=0}^{N} \hat{f}(\tau = t - p_l x_k^2, p_l), \tag{40}$$

where $p_l = l\Delta p$ and $x_k = k\Delta x$.

Parabolic Radon

We now want to perform a Fourier transform over the time variable

$$f(x_k, \omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt \sum_{l=0}^{N} \hat{f}(\tau = t - p_l x_k^2, p_l) \exp(-i\omega t), \quad (41)$$

which becomes after substitution of variable $u = t - p_l x_l^2$

$$f(x_k, \omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} du \sum_{l=0}^{N} \hat{f}(\tau = u, p_l) \exp[-i\omega(u + p_l x_k^2)]. \quad (42)$$

The last equation is then

$$f(x_k, \omega) = \sum_{l=0}^{N} \hat{F}(\omega, p_l) \exp[-i\omega p_l x_k^2]. \quad (43)$$

Parabolic radon

To compute the forward transform we consider the $\hat{F}(\omega, p_l)$ as unknowns, and solve the linear system of equations given in (43). This can easily be done by writing equation (43) as a matrix equation and using the least-squares method.

$$\mathbf{f}(\omega) = \mathbf{L}\hat{\mathbf{F}}(\omega) \tag{44}$$

where $\mathbf{f}$ and $hat\mathbf{F}$ are vectors with elements $f_k = f(x_k, \omega)$ and $\hat{F}_k = \hat{F}(x_k)$. The matrix $\mathbf{L}$ have elements $L_{lk} = \exp[-i\omega p_l x_k^2$.

Parabolic radon

After muting of $\hat{\mathbf{F}}$ to remove multiples we solve the equation

$$\hat{\mathbf{F}}(\omega) = \mathbf{L}^{-1}\mathbf{f}(\omega) \tag{45}$$

with respect to $\mathbf{f}$ to compute the inverse transform.

Parabolic Radon

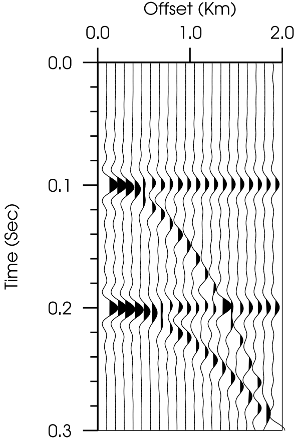


Figure: Input data with primary and multiple reflections. A normal moveout correction has been applied.

Parabolic Radon

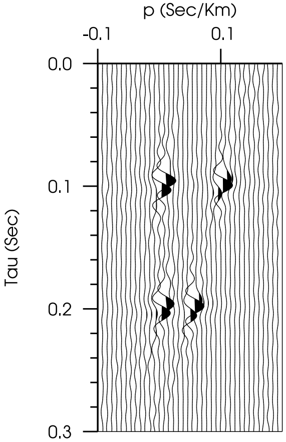


Figure: The parabolic radon transform of the data shown in figure 7.

Parabolic Radon

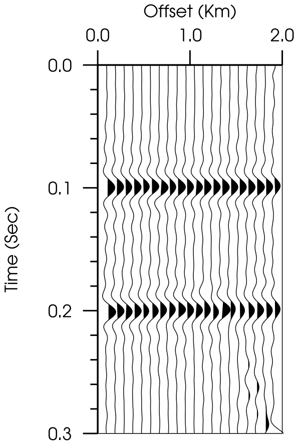


Figure: Primary reflections estimated from the input data in figure 7 using the parabolic radon transform.